Wave speed

antocciphysics | TEACHER KEY | CPA

Learning target: Use speed, frequency and wavelength in a consistent relationship.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 5, Day 12; 3 course days later.*
+40 J; internal energy increases.

02. Explain a worked example (worked)
Two cycles of 3 m versus three cycles of 2 m; speed remains 6 m/s.

03. Look and explain (model)
Higher frequency has half the wavelength.

04. Build the solution (practice)
8 m/s; 1 m.

05. Use the idea (reason)
Hz/m=1/(s·m), not m/s; correct product fλ gives m/s.

06. Choose a method (mixed) *Review: Unit 5, Day 3; 12 course days later.*
Electromagnetic radiation can travel in a vacuum; conduction/convection require matter.

07. Exit check (exit)
15 m/s.

08. STRETCH: Apply it another way (stretch)
Examples (1 Hz, 12 m),(2 Hz, 6 m),(4 Hz, 3 m).

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01. Bring back an earlier idea (retrieve) *Review: Unit 5, Day 12; 3 course days later.*
Carnot limit $\eta \approx 0.4$; $W_{max} \approx 280$ J. Irreversibility and losses reduce actual useful work.

02. Explain a worked example (worked)
Two cycles of 3 m versus three cycles of 2 m; speed remains 6 m/s.

03. Represent and interpret (model)
Higher frequency has half the wavelength.

04. Solve and justify (practice)
λ=.80 m; f 680 Hz.

05. Analyze the reasoning (reason)
Hz/m=1/(s·m), not m/s; correct product fλ gives m/s.

06. Choose a method (mixed) *Review: Unit 5, Day 3; 12 course days later.*
Net thermal radiation transfers energy to the cooler radiative environment. Absolute zero is not required.

07. Exit check (exit)
30 m/s;.60 m.

08. STRETCH: Apply it another way (stretch)
3 and 2 m; driven oscillations keep the source frequency across a stationary boundary.